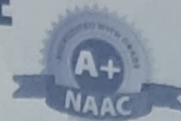




St. MARTIN'S ENGINEERING COLLEGE

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ASSIGNMENT ANSWER BOOKLET

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Course : B-Tech

Branch : CSE

Year : I/II/III/IV

Sem : I / II

Assignment : I / II

Subject Name : Computer Networks

Last Date of Submission: 31/8/26

MARKS AWARDED

Q.No.	a	b	Total
1			
2			
3			
4			
5			
Total			

Actual Date of Submission : 27/8/26

Signature of the Student

Signature of the Evaluator

Start Writing from Here

1. Explain layered architecture of OSI reference model.

Layered Architecture of OSI Reference Model:

The OSI is a conceptual model developed by ISO to describe how data is communicated between two systems. It consists of 7 layers, where each layer performs a specific function.

1. Physical layer (layer 1):

This is the lowest layer. It is responsible for actual physical connection between the devices. This contains information in the form



of bits. Transmits individual bits from one node to the next. When receiving data, this layer will get the signal received & convert it into 0's & 1's and send them to the data link layer, which will put the frame back together.

2) Data link layer (layer 2):

This is responsible for the node to node delivery of the message. The main function is to make sure data transfer is error-free from one node to another, over the physical layer. When a packet arrives in a network, it is the responsibility of DLL to transmit to the Host using its MAC address.

3) Network layer (layer 3):

It works for the transmission of data from one host to the other located in different networks. It also takes care of packet routing, selection of the shortest path to transmit the packet, from the number of routes available. The sender & receiver's IP addresses are placed in the header by the network layer.

4) Transport layer (layer 4):

This provides services to the application layer & takes service from the network layer. The data, here, is referred to as segments. It is responsible for the End-to-End delivery of the complete message.



5) Session layer (layer 5):

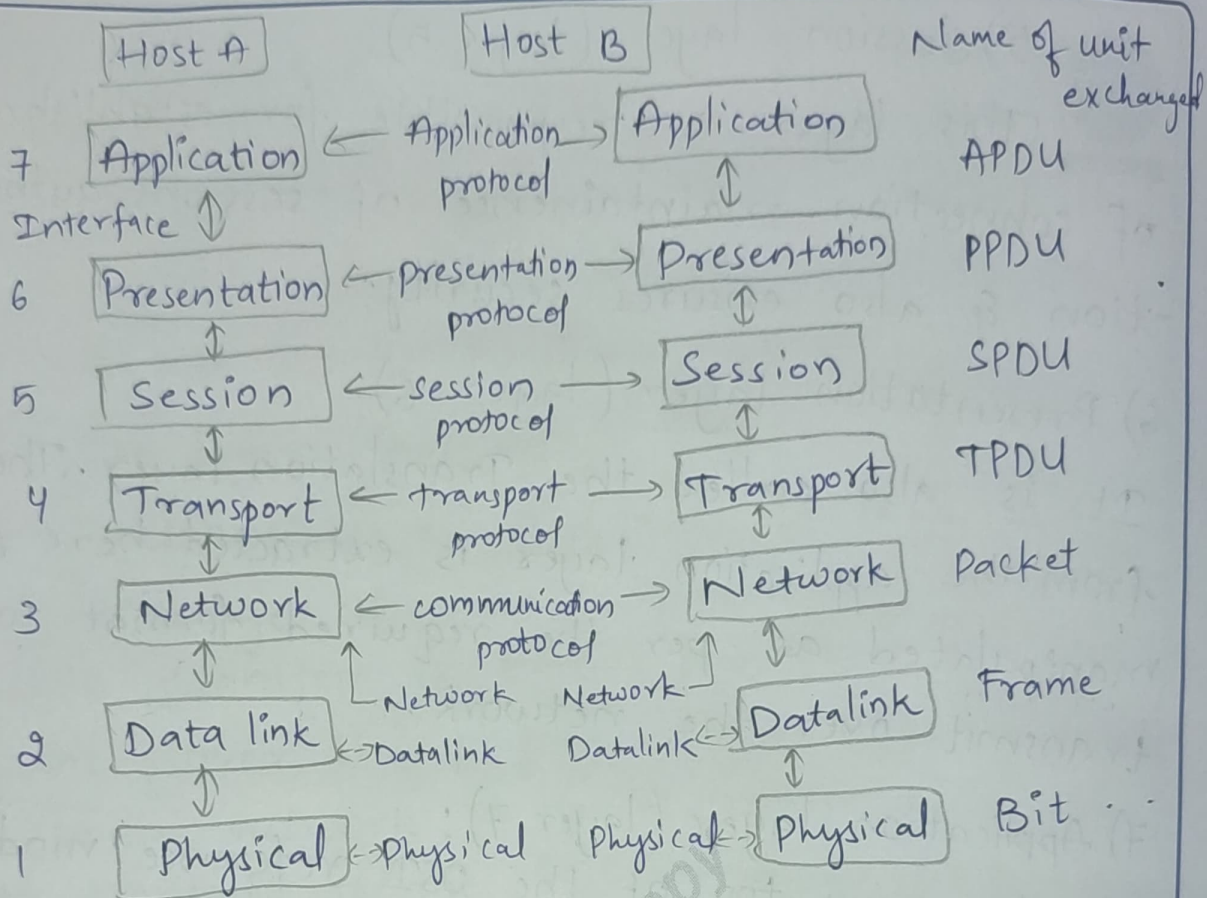
This layer is responsible for establishment of connection maintenance of sessions, authentication & also ensures security.

6) Presentation layer (layer 6):

It is also called the Translation layer. The data from the application layer is extracted here & manipulated as per the required format to transmit over the network.

7) Application layer (layer 7):

At the very top of the OSI reference model stack of layers, we find application layer which is implemented by the network applications. This layer also serves as a window for the application services to access the network & for the application services to access the network & for displaying the received information to the user.



Internet subnet protocol.

2 Explain error detection and error detection techniques used at the data link layer.

Data link layer (DLL):

The data link layer is responsible for the node to node delivery of the message. The main function is to make sure data transfer is error free from one node to another, over the physical layer. When a packet arrives in a network, it is the responsibility of DLL to transmit it to the host using its MAC address.

Error detection:

It is a mechanism used by the Data link layer to detect whether the data received by the receiver is corrupted or contains errors during



transmission.

The data link layer adds some extra redundant bits to the original data before transmission. The receiver uses these bits to check whether an error has occurred.

Error detection techniques:

The main techniques are:

1. Parity check:

- A parity bit is added to the data, it can be even parity (or) odd parity.
- The receiver checks the parity of the received data.
- If the parity is different from the expected value, an error is detected.

ex: Data = 1011001

For even parity, the number of 1's is made even by adding a parity bit.

2. Check sum:

- The data is divided into fixed-size blocks.
- The blocks are added using binary addition.
- The complement of the sum is called the checksum.
- The checksum is sent along with the data.
- At the receiver the data is recalculated & compared with the received checksum.
- If they do not match, an error is detected.

3. Cyclic Redundancy Check (CRC):

- This is a powerful error-detection technique.



- The data bits are treated as a binary number & divided by a predefined generator polynomial.
- The remainder obtained from the division is called the CRC.
- This remainder is attached to the original data & transmitted.
- At the receiver, the same division is performed.
- If the remainder is zero, the data is assumed to be error-free, otherwise, an error is detected.

Advantage:

very effective in detecting burst errors & widely used in computer networks.

3. What are the elementary data link layer protocols? Discuss.

→ The Data link layer is responsible for reliable communication between two directly connected nodes.

→ The protocols are:

A) Protocols for noiseless channels:

These assume that the channel is error-free

1. Simplest protocol:

→ It has no flow control & no error control.

→ Data frames travel only in one direction, from sender to receiver.

→ The sender sends frames whenever data is available.



→ The receiver immediately accepts the frame and passes the data to the network layer.

→ It assumes the receiver can process the frames fast enough.

B) Protocols for Noisy channels:

These are used when frames may be lost (or) damaged during transmission.

1. Stop-and-wait ARQ:

→ It adds error control to the stop-and-wait protocol.

→ The sender sends a frame and waits for an ACK.

→ If the frame is damaged, it is discarded by the receiver.

→ If the ACK is not received within the required time, the sender retransmits the frame.

→ Sequence numbers are used to identify duplicate frames.

2. Go-Back-N ARQ:

→ Multiple frames can be sent before receiving acknowledgements.

→ This is called pipelining.

→ Frames are numbered using sequence numbers.

→ If one frame is lost or damaged, the sender goes back and retransmits that frame & all subsequent frames.

→ It uses a sliding window.



Ex:- suppose frames!

0 1 2 3 4

gf frame 2 is lost:

0✓ 1✓ 2× 3 4

The sender retransmits:

2→3→4

3. Selective Repeat ARQ:

Multiple frames can also sent before receiving ACKs.

→ It uses sliding window.

→ If one frame is lost (or) damaged, only that particular frame is retransmitted.

→ The receiver can temporarily store correctly received frames that arrive out of order.

→ It is more efficient than Go-Back-N on a noisy channel.

Ex: 0✓ 1✓ 2× 3✓ 4✓

only frame 2 is retransmitted:

2→

4. Explain CSMA protocols & discuss how carrier sensing helps reduce collisions.

CSMA (Carrier sense multiple Access):

It is a carrier sense multiple access based on media access protocol to sense the traffic on a channel before transmitting the data. It means



that if the channel is idle, the station can send data to the channel. Otherwise, it must wait until the channel becomes idle. Hence, it reduces the chances of a collision on a transmission medium.

Protocols:-

CSMA/CD: It is a carrier sense multiple access/collision detection network protocol to transmit data frames. The CSMA/CD protocol works with a medium access control layer. Therefore, it first senses the shared channel before broadcasting the frames & if the channel is idle, it transmits a frame to check whether the transmission was successful. If the frame is successfully received, the station sends another frame. If any collision is detected in the CSMA/CD, the station sends a jam (stop signal) to the shared channel to terminate data transmission. After that it waits for a random time before sending a frame to a channel.

CSMA/CA: It is a carrier sense multiple access/collision avoidance network protocol for carrier transmission of data frames. It is a protocol that works with a medium access control layer. When a data frame is sent to a channel, it receives an acknowledgement to check whether the channel is clear. If the station receives only a single (own) acknowledgement that means the data



frame has been successfully transmitted to the receiver. But if it gets two signals (its own & one more) a collision of the frame occurs in the shared channel. Detects the collision of the frame when a sender receives an acknowledgement signal.

Collision free protocols:

- Bitmap protocol: A collision-free protocol in which each station is assigned a bit, & stations with data set their bits to 1 as to indicate that they want to transmit.

- Binary countdown protocol.

A protocol in which stations used their binary addresses to compete for the channel, and the station with the highest-priority binary address gets access.

- Limited contention protocol.

A protocol in which stations used their binary addresses to compete for the channel & the station with the highest-priority binary

→ A protocol in which only a limited numbers of stations compete for the channel at a time to reduce collisions.

- Adaptive tree walk protocol: A protocol that uses a binary tree to divide competing stations into smaller groups after a collision until a



station can transmit successfully.

5 Explain flooding as a routing algorithm. Discuss its advantages & limitations.

Routing Algorithms:

It is that part of the network layer software responsible for deciding which output line an incoming packet should be transmitted on. If the subnet uses datagrams internally, this decision must be made anew for every arriving data packet.

Flooding as a routing algorithm:

→ It is a static routing algorithm in which every incoming packet is sent through all outgoing links except the link from which the packet arrived. Thus, multiple copies of the same packet travel through different paths in the network.

→ One such measure is to have a hop counter contained in the header of each packet, which is decremented at each hop, with the packet being discarded when the counter reaches zero. Ideally, the hop counter should be initialized to the length of the path from source to destination.

Advantages of flooding:

1. Simple: The algorithm is easy to understand and implement.



2. Highly robust:

Since packets are sent through many paths, the failure of one (or) more links does not necessarily prevent delivery.

3. Shortest path:

It always finds a shortest path because all possible paths are explored simultaneously.

4. Reliable:

There is a high probability that at least one copy of the packet will reach the destination.

Limitations:

1. Large number of Duplicate packets.

The same packets can reach a router through many different paths.

2. High Bandwidth consumption:

A large number of copies occupy network links & waste bandwidth.

3. Network Congestion:

Excessive duplicate packets as can create congestion & reduce network performance.

4. Processing overhead:

Routers have to process & forward many copies of the same packet.

5. Requires control mechanisms:

Hop counters, sequence numbers, or other methods are needed to prevent packets from continuing indefinitely.